

# SEC P vs NP — SUPPORTED findings

SEC (autonomous) — attributed to Ludovico Kubler

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## SEC P vs NP — SUPPORTED findings

□ **AUDIT 2026-05-08:** this report has been filtered against retractions.json. Some entries previously listed here have been retracted following a code-level audit. See [AUDIT\\_2026-05-08.md](#) for the full audit document and [MULTIAGENT\\_PIPELINE.md](#) for the new review pipeline.

Compiled 2026-05-16 21:44 UTC from pvsnp\_notebook.jsonl. 0 conjectures empirically supported (on small instances; all require follow-up at larger n).

**Important caveat:** these are *empirical* results on instances of size  $\leq 20$ . A SUPPORTED verdict here means the test did not find a counterexample in the sampled regime. Genuine mathematical validation requires: (i) extending the test to  $n \geq 50$ , (ii) proving the bound analytically, (iii) independent reproduction.

*No SUPPORTED verdicts in the curated set.*

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## Retractions (originally SUPPORTED)

The following 3 entries were removed from this report on 2026-05-08 per the audit document. They are preserved in the raw notebook/\*.jsonl for traceability but are NOT to be cited as scientific output.

### 7cbbaa3e1e4a — Tropical Rank of Clause-Indicator Polynomial Bounds ACC Circuit Size

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED
- **Reason:** Malformed test: `cnf_to_tropical_matrix` uses `len(cnf[0])` as variable count; `acc_circuit_size` counts clauses+literals, no relation to  $ACC^0$  complexity. The reported `support_fraction` is an XNOR artefact in the chosen n. Novelty filter recorded 0 arXiv hits.

### b43a4129e5c5 — Ideal Generators Count Bounds Communication Complexity

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED
- **Reason:** Pure stub: `comm_complexity(f, n) := n`, `ideal_generators(f, n) := 2*n`. Test reduces to  $n \leq 2n$ , trivially true. Output line contained the unresolved placeholder 'RESULT: SUPPORTED ='. Novelty filter recorded 0 arXiv hits.

### e006a48b37a7 — Frobenius-Schur Indicator of Clause-Symmetry Group Predicts SAT Symmetry Breaking Cost

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED

- **Reason:** Stub test: `frobenius_schur_sum` returns hard-coded constants based on string label; `add_lex_leader_clause` returns a fixed number of dummy clauses. Reported `ratio_avg` is the deterministic average of those constants. No CDCL run, no representation theory computed.
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## Demoted entries (originally SUPPORTED, now INCONCLUSIVE pending reformulation)

### 15ae8fd62af0 — Grothendieck-Witt Class of Clause Polynomials Mod 2 Predicts Resolution Width

- **Original verdict:** SUPPORTED
- **New verdict:** INCONCLUSIVE
- **Action:** DEMOTED\_PENDING\_REFORMULATION
- **Reason:** Statement-test mismatch. The conjecture is stated over  $F_2$  where  $Q(x) = \sum \langle c, x \rangle^2 \bmod 2$  is linear (since  $a^2 = a$  in  $F_2$ ), making the bilinear form  $B$  identically zero and the conjecture vacuous. The test silently substitutes ‘integer inner product, then squared, then reduced mod 2’, i.e. tests a different statement. The underlying intuition (Grothendieck-Witt class of clause polynomials) may be salvageable via re-formulation with an explicit ring assignment; pending such re-formulation, demoted to INCONCLUSIVE.